SIMATS ENGINEERING

ASSESSMENT TOOL 1: INDUSTRY PROBLEM SOLVING TASKS

COURSE NAME:	Programming in Java	**COURSE CODE:**	CSA0915- SLOT B
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Assessment Tasks

1. Industry Scenario:
   A hospital wants to develop a desktop-based appointment management system for its reception counter.

Question:
Design and develop a Java AWT application that allows the receptionist to register a patient appointment. The application should provide fields for Patient ID, Patient Name, Age, Gender, Department, Doctor Name, and Appointment Date. Use appropriate AWT components such as Label, TextField, CheckboxGroup, Choice, Button, and Frame.
When the Book Appointment button is clicked, validate the entered information and generate an appointment summary containing the patient details, selected department, doctor, and appointment date. Provide a Clear button to reset all fields.

Requirements:

1. Use suitable AWT components and a container.
2. Use an appropriate layout.
3. Handle button events.
4. Validate mandatory fields.
5. Display the appointment confirmation.
6. Provide Clear functionality.

Test Cases:

tc1_input- P101, Ravi, 35, Male, Cardiology, Dr. Kumar
tc1_output- Appointment Booked Successfully
tc2_input—P102, Priya, 28, Female, Neurology, Dr. Meena
tc2_output- Appointment Booked Successfully
tc3_input- Patient Name empty
tc3_output- Patient Name Required
tc4_input- Patient ID empty
tc4_output- Patient ID Required
tc5_input-Age = 0
tc5_output-Invalid Age

CODE:

```
import java.awt.*;
import java.awt.event.*;

public class HospitalAppointment extends Frame implements ActionListener {

    Label patientIdLabel, nameLabel, ageLabel, genderLabel;
    Label departmentLabel, doctorLabel, dateLabel, summaryLabel;

    TextField patientIdField, nameField, ageField, dateField;

    Checkbox male, female, other;
    CheckboxGroup genderGroup;

    Choice departmentChoice, doctorChoice;

    Button bookButton, clearButton;

    TextArea summaryArea;

    public HospitalAppointment() {

        setTitle("Hospital Appointment Management System");
        setSize(650, 550);
        setLayout(new BorderLayout(10, 10));

        Label title = new Label("HOSPITAL APPOINTMENT MANAGEMENT SYSTEM",
                                Label.CENTER);
        title.setFont(new Font("Arial", Font.BOLD, 20));
        add(title, BorderLayout.NORTH);


        Panel formPanel = new Panel();
        formPanel.setLayout(new GridLayout(7, 2, 10, 10));


        patientIdLabel = new Label("Patient ID:");
        patientIdField = new TextField();

        formPanel.add(patientIdLabel);
        formPanel.add(patientIdField);

        nameLabel = new Label("Patient Name:");
        nameField = new TextField();
```

```
        formPanel.add(nameLabel);
        formPanel.add(nameField);


        ageLabel = new Label("Age:");
        ageField = new TextField();

        formPanel.add(ageLabel);
        formPanel.add(ageField);


        genderLabel = new Label("Gender:");

        Panel genderPanel = new Panel(new FlowLayout(FlowLayout.LEFT));

        genderGroup = new CheckboxGroup();

        male = new Checkbox("Male", genderGroup, false);
        female = new Checkbox("Female", genderGroup, false);
        other = new Checkbox("Other", genderGroup, false);

        genderPanel.add(male);
        genderPanel.add(female);
        genderPanel.add(other);

        formPanel.add(genderLabel);
        formPanel.add(genderPanel);


        departmentLabel = new Label("Department:");
        departmentChoice = new Choice();

        departmentChoice.add("Select Department");
        departmentChoice.add("Cardiology");
        departmentChoice.add("Neurology");
        departmentChoice.add("Orthopedics");
        departmentChoice.add("Pediatrics");
        departmentChoice.add("Dermatology");

        formPanel.add(departmentLabel);
        formPanel.add(departmentChoice);


        doctorLabel = new Label("Doctor Name:");
```

```
        doctorChoice = new Choice();

        doctorChoice.add("Select Doctor");
        doctorChoice.add("Dr. Kumar");
        doctorChoice.add("Dr. Meena");
        doctorChoice.add("Dr. Raj");
        doctorChoice.add("Dr. Priya");

        formPanel.add(doctorLabel);
        formPanel.add(doctorChoice);


        dateLabel = new Label("Appointment Date:");
        dateField = new TextField();

        formPanel.add(dateLabel);
        formPanel.add(dateField);

        add(formPanel, BorderLayout.CENTER);


        Panel bottomPanel = new Panel(new BorderLayout(5, 5));

        Panel buttonPanel = new Panel();

        bookButton = new Button("Book Appointment");
        clearButton = new Button("Clear");

        bookButton.addActionListener(this);
        clearButton.addActionListener(this);

        buttonPanel.add(bookButton);
        buttonPanel.add(clearButton);

        bottomPanel.add(buttonPanel, BorderLayout.NORTH);

        summaryLabel = new Label("Appointment Summary:");
        bottomPanel.add(summaryLabel, BorderLayout.CENTER);

        summaryArea = new TextArea("", 8, 60,
                                   TextArea.SCROLLBARS_VERTICAL_ONLY);
        summaryArea.setEditable(false);

        bottomPanel.add(summaryArea, BorderLayout.SOUTH);
```

```
        add(bottomPanel, BorderLayout.SOUTH);


        addWindowListener(new WindowAdapter() {
            public void windowClosing(WindowEvent e) {
                dispose();
            }
        });

        setVisible(true);
    }
    public void actionPerformed(ActionEvent e) {


        if (e.getSource() == bookButton) {
            bookAppointment();
        }

        if (e.getSource() == clearButton) {
            clearFields();
        }
    }


    private void bookAppointment() {

        String patientId = patientIdField.getText().trim();
        String patientName = nameField.getText().trim();
        String ageText = ageField.getText().trim();
        String date = dateField.getText().trim();


        if (patientId.isEmpty()) {
            summaryArea.setText("Patient ID Required");
            return;
        }


        if (patientName.isEmpty()) {
            summaryArea.setText("Patient Name Required");
            return;
        }


        if (ageText.isEmpty()) {
```

```
            summaryArea.setText("Age Required");
            return;
        }

        int age;

        try {
            age = Integer.parseInt(ageText);

            if (age <= 0 || age > 120) {
                summaryArea.setText("Invalid Age");
                return;
            }

        } catch (NumberFormatException ex) {
            summaryArea.setText("Invalid Age");
            return;
        }


        Checkbox selectedGender = genderGroup.getSelectedCheckbox();

        if (selectedGender == null) {
            summaryArea.setText("Gender Required");
            return;
        }


        if (departmentChoice.getSelectedIndex() == 0) {
            summaryArea.setText("Department Required");
            return;
        }


        if (doctorChoice.getSelectedIndex() == 0) {
            summaryArea.setText("Doctor Name Required");
            return;
        }


        if (date.isEmpty()) {
            summaryArea.setText("Appointment Date Required");
            return;
        }
```

```
        String gender = selectedGender.getLabel();
        String department = departmentChoice.getSelectedItem();
        String doctor = doctorChoice.getSelectedItem();


        summaryArea.setText(
            "Appointment Booked Successfully\n\n" +
            "Patient ID       : " + patientId + "\n" +
            "Patient Name     : " + patientName + "\n" +
            "Age              : " + age + "\n" +
            "Gender           : " + gender + "\n" +
            "Department       : " + department + "\n" +
            "Doctor Name      : " + doctor + "\n" +
            "Appointment Date : " + date
        );
    }


    private void clearFields() {

        patientIdField.setText("");
        nameField.setText("");
        ageField.setText("");
        dateField.setText("");

        genderGroup.setSelectedCheckbox(null);

        departmentChoice.select(0);
        doctorChoice.select(0);

        summaryArea.setText("");
    }


    public static void main(String[] args)
 {
        new HospitalAppointment();
    }
}
```